



Object detection

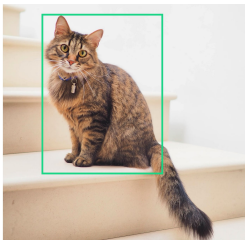
Laboratory of robotics and control systems
ITT PROJECT MANAGEMENT SERVICES L.L.C

CV tasks

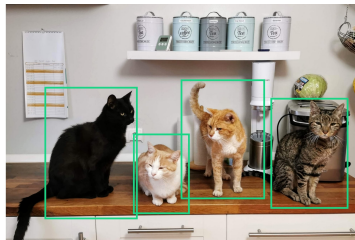


YES

classification



classification + localisation



object detection

Problem statement

Objective:

- Localise the object
- Define the class of the object

Contains box description and object class

$$(x_i, y_i, w_i, h_i, c_i)$$

Input:

- An image

Output:

- One predictions (**Classification + Localisation**)
- Set of predictions (**Object detection**)

Objects and materials

Объекты (Things)

- Имеют определенные размер и формы



Найти «человека»

Материалы (Stuff)

- Однородный или повторяющийся шаблон мелких деталей
- Нет определенного размера и формы



Найти «дорогу»

Take another slide

Multi-class

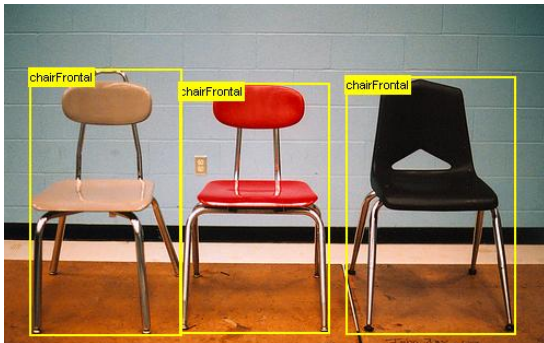
- Pascal Visual Object Classes
- Microsoft COCO

Single-class

- Faces
- Cars
- Pedestrians
- Road signs

Pascal VOC

- 20 классов
- 11,530 изображений
- 27,450 регионов интересов (RoI)
- 6,929 сегментационных масок



<http://host.robots.ox.ac.uk/pascal/VOC/>

Microsoft COCO

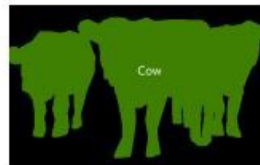
- 80 классов
- >200K изображений (train, val, test)
- 1.5M экземпляров объектов
- 500K выделенных объектов (с масками)



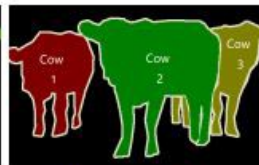
(a) Image Classification



(b) Object Detection



(c) Semantic Segmentation



(d) Instance Segmentation

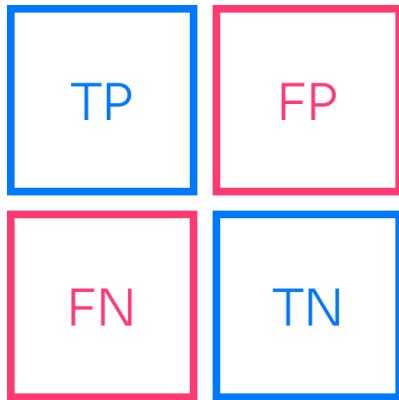
<http://cocodataset.org/home>

Quality Metrics

- Intersection over Union (IoU)
- Precision and Recall
- AUC, ROC
- Average Precision (AP), Mean AP (mAP)
- Miss rate
- FPPI (False Positive per image)

How to count TP, FP, TN, FN

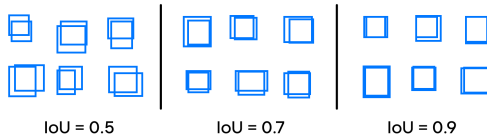
- Through the whole test data
- For each image



IoU - Intersection over Union

$$\text{Score} = \frac{\text{Area of overlap}}{\text{Area of union}}$$


Detection, if $\text{IoU} > p$ (approx. : 0.5)

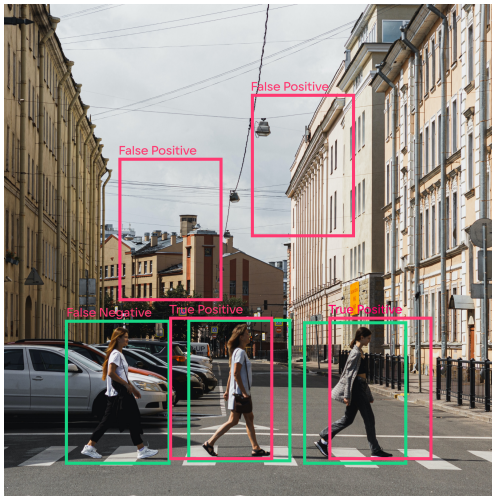


Precision and Recall

$$precision = \frac{TP}{TP + FP}$$

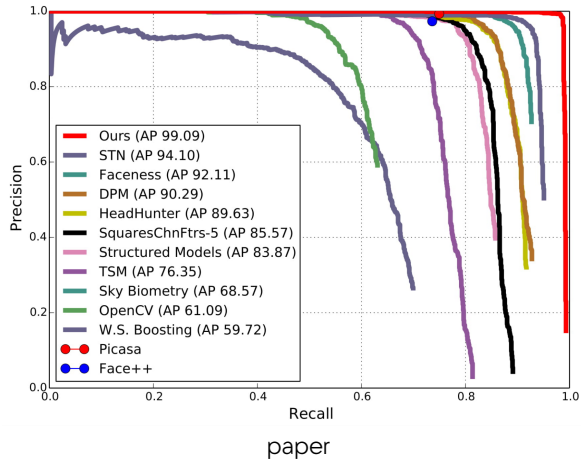
$$recall = \frac{TP}{TP + FN}$$

- $IoU > p \rightarrow TP$
- $IoU < p \rightarrow FP$
- missed object $\rightarrow FN$
- p affects recall and precision values
- These metrics are used together



Precision-Recall curve

- visualize tradeoff between precision and recall metrics
- Low Recall and High Precision - avoid FPs but skip objects (lots of FN)
- Low Precision and High Recall - does not skip objects (no FNs) but predicts excessive objects (lots of FP)
- usefull in tasks with imbalanced classes
- **AUC** - the area under the PR curve



AP, mAP

1 class

$$AP = \frac{1}{n} \sum_{r \in \{0, 0.1, \dots, 1\}} p_{interp}(r)$$

$p_{interp}(r)$ - maximum precision for recall is more than r

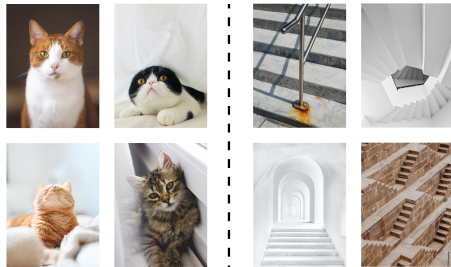
N classes

- mAP - the average AP by class
- $mAP@0.5$ - the average AP by class with IoU threshold = 0.5
- $mAP@0.5 : 0.95$ - the average mAP by IoU thresholds in $[0.5, 0.95]$ with step = 0.05

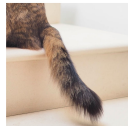
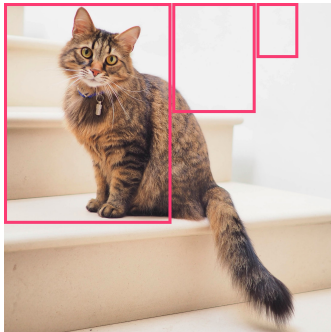
Sliding window



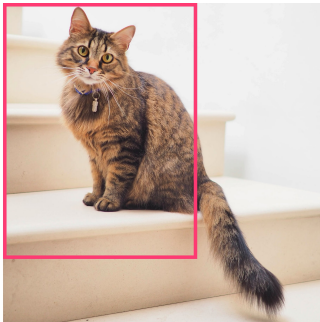
Training Data



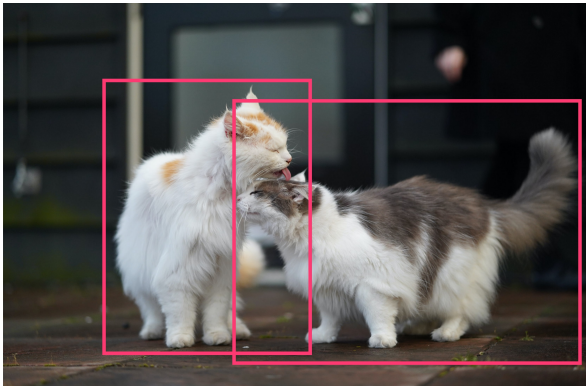
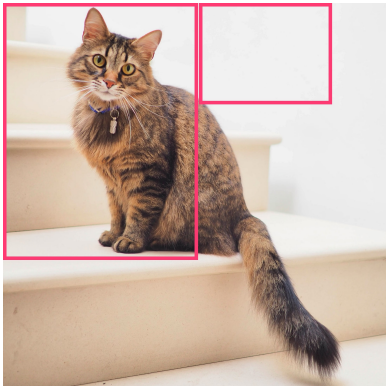
Varying of the window size



Varying of the picture size



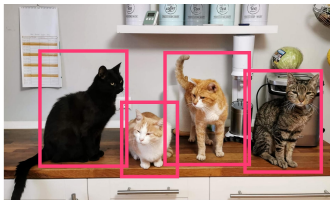
Varying of the window aspect ratio



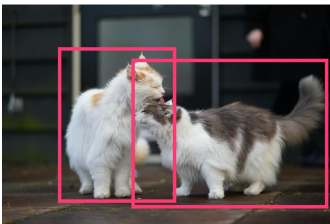
Problems



Different sizes



Different proportions



Overlaps and difficult shapes



Multiple responses

Dealing with multiple responses

Meaning

1. Clusterize bbs
2. Take average coordinates in cluster.

NMS (non max supression)

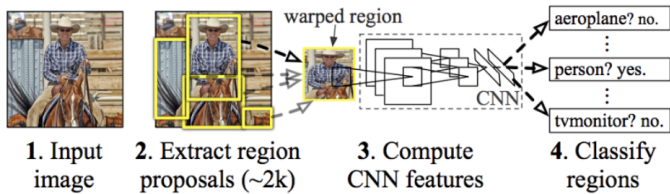
1. Sort bounding boxes by their confidence score.
2. Calculate IoU for the ground true and sliding windows.
3. Drop windows with metric value lower than threshold.
4. Continue until convergence will be achieved.

Results

- Can not be implemented inside NN
- Post-processing

Region-based Convolution Networks

R-CNN: *Regions with CNN features*



- two-stage detector
- Find region proposals with classic algorithm
- For each region proposal Alexnet-like feature extractor is applied
- SVMs as classifiers
- linear regression model for bbs refinements
- Paper

Region Proposals



Figure 2: Two examples of our selective search showing the necessity of different scales. On the left we find many objects at different scales. On the right we necessarily find the objects at different scales as the girl is contained by the tv.

- Find about 2000 region proposals
- different size
- different aspect of ratio
- Paper

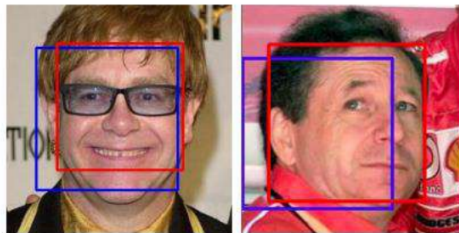
After Getting Feature vector

Bbox clarification

- input - feature map from CNN
- output - 4 parameters (dx, dy, dw, dh)
- linear regression model

Region classification

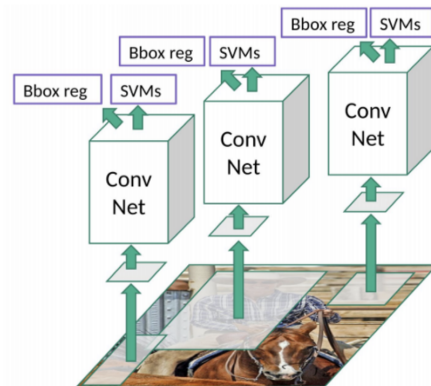
- input - feature map from CNN
- output - probabilities of having each class
- model consists of N linear SVM



General structure

Network R-CNN can be divided into steps:

1. Selection of candidate regions
2. Converting regions to one size
3. Embedding
4. N binary classification
5. Additional processing of the bounding boxes



R-CNN

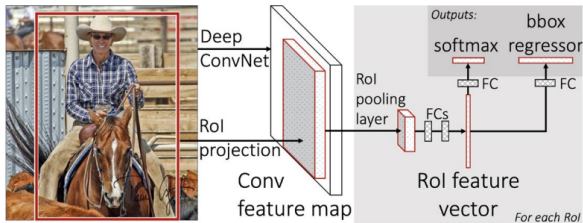
RCNN to Fast-RCNN

Problems:

- Selective Search works too long
- The need to scale hypotheses
- Extra calculations when calculating features for each hypothesis
- Complex training procedure (extractor, bbs regression and SVMs are trained separately)

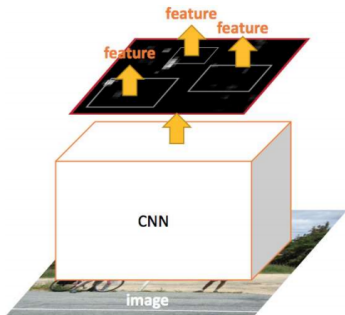
Solution:

- Calculating the feature map of the entire image
- No need to scale hypotheses
- All modules are trained collectively
- Training with 2 types of outputs: object type classifier, bbox regressor for **each** hypothesis of selective search



ROI-Pooling

- ROI = Region of Interest
- Calculating ROIs (features) once for the whole image
- Reduce ROI size by ROI-pooling
- Drop to FCs



0,2	0,9	0,4	0,1	0,1	0,2
0,1	0,8	0,4	0,5	0,9	0,4
0,2	0,6	0,5	0,8	0,5	0,5
0,9	0,4	0,4	0,2	0,5	0,4
0,9	0,2	0,1	0,2	0,2	0,5
0,6	0,5	0,4	0,2	0,4	0,6

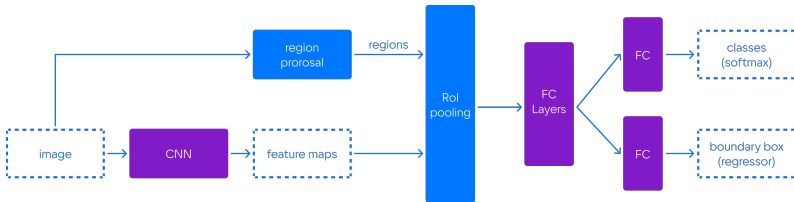
0,9	0,8
0,9	0,4

0,2	0,9	0,4	0,1	0,1	0,2
0,1	0,8	0,4	0,5	0,9	0,4
0,2	0,6	0,5	0,8	0,5	0,5
0,9	0,4	0,4	0,2	0,5	0,4
0,9	0,2	0,1	0,2	0,2	0,5
0,6	0,5	0,4	0,2	0,4	0,6

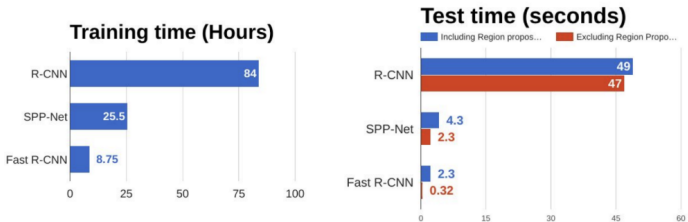
0,2	0,9	0,4	0,1	0,1	0,2
0,1	0,8	0,4	0,5	0,9	0,4
0,2	0,6	0,5	0,8	0,5	0,5
0,9	0,4	0,4	0,2	0,5	0,4
0,9	0,2	0,1	0,2	0,2	0,5
0,6	0,5	0,4	0,2	0,4	0,6

Fast R-CNN

Architecture



Benchmarks



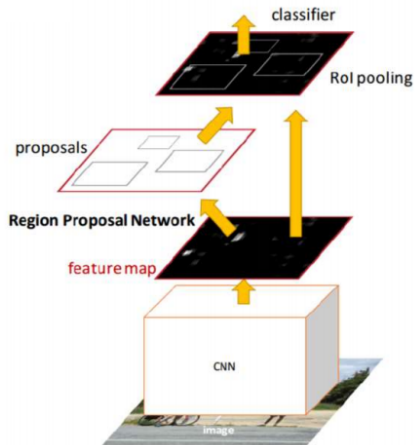
Fast R-CNN to Faster R-CNN

Problems:

- Selective Search works too long

Solution:

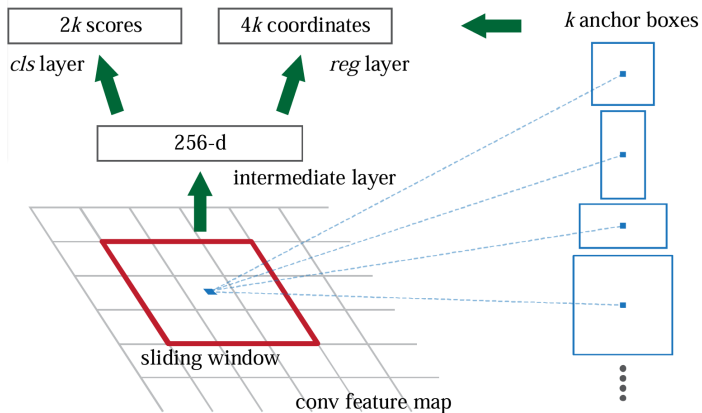
- **RPN** - Region Proposal Network
- Faster R-CNN = FAST R-CNN + RPN
- The dependence on the external hypothesis generation algorithm is eliminated
- Object detection is done in one pass by one network



<https://evogeek.ru/articles/615503/>

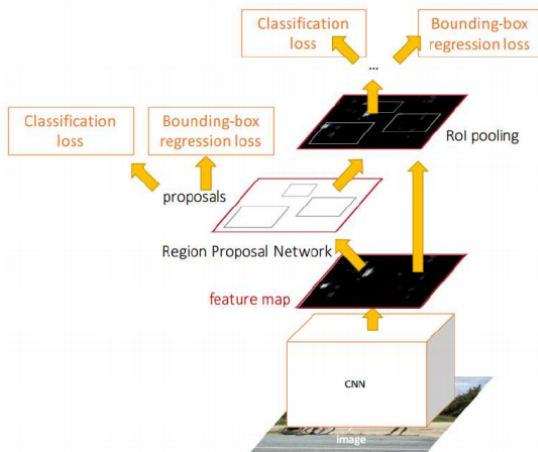
Region Proposal Network (RPN)

- Idea of anchor boxes.
- sliding window.
- 9 anchor boxes in each window.
- Then anchor boxes are refined.



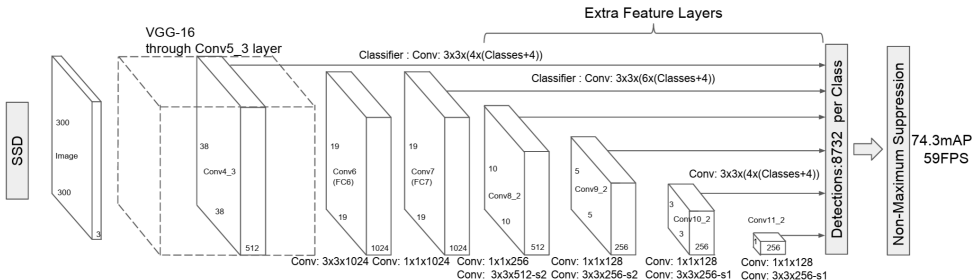
Training

- One network, four outputs
- Train collectively in multi-task mode



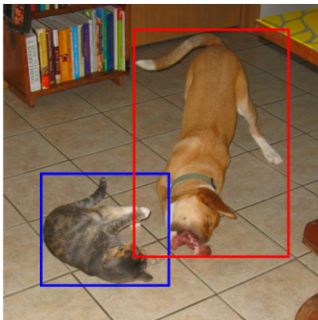
<https://slides.com/yunqy/deck-4/fullscreen>

SSD (Single Shot Detector)

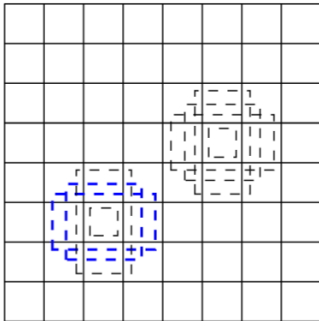


- single-shot detector
- Base network - VGG
- Add convolution layers to base network
- Predict offsets to default boxes of different scales and aspect ratios
- Predict anchor boxes associated confidences
- Multiple output layers at different resolutions

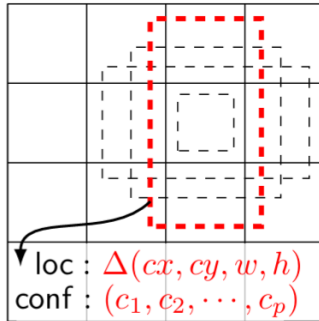
SSD Predictions



(a) Image with GT boxes



(b) 8×8 feature map



loc : $\Delta(cx, cy, w, h)$
 conf : $(c_1, c_2, \dots, c_p)$

(c) 4×4 feature map

- TP boxes - default boxes matching to any ground truth with IoU > 0.5

$$L(x, c, l, g) = \frac{1}{N} (L_{conf}(x, c) + \alpha L_{loc}(x, l, g))$$

- multiple classes confidences (c), predicted box (l), ground truth box (g),

SSD Loss

$$L(x, c, l, g) = \frac{1}{N}(L_{conf}(x, c) + \alpha L_{loc}(x, l, g))$$

Designations:

- $x_{ij}^k = \{1, 0\}$ - an indicator for matching the i-th default box to the j-th ground truth box of category p
- d - default box
- $\hat{c}_i^0$ - confidence of background class

$$L_{loc}(x, l, g) = \sum_{i \in Pos} \sum_{m \in \{cx, cy, w, h\}}^N x_{ij}^k \text{smooth}_{L1}(l_i^m - \hat{g}_j^m)$$

$$\hat{g}_j^{cx} = (g_j^{cx} - d_i^{cx})/d_i^w \quad \hat{g}_j^{cy} = (g_j^{cy} - d_i^{cy})/d_i^h \quad \hat{g}_j^w = \log\left(\frac{g_j^w}{d_i^w}\right) \quad \hat{g}_j^h = \log\left(\frac{g_j^h}{d_i^h}\right)$$

SSD Loss

$$L(x, c, l, g) = \frac{1}{N}(L_{conf}(x, c) + \alpha L_{loc}(x, l, g))$$

$$L_{conf}(x, c) = - \sum_{i \in Pos}^N x_{ij}^p \log(\hat{c}_i^p) - \sum_{i \in Neg} \log(\hat{c}_i^0) \quad \text{where} \quad \hat{c}_i^p = \frac{\exp(c_i^p)}{\sum_p \exp(c_i^p)}$$

To sum up

- FN boxes are fined by L_{conf}
- FP boxes are fined by L_{conf} and L_{loc}
- α is set to 1 by cross validation

SSD Features

- one shot detection (faster and more accurate than YOLO)
- fixed number of bbs from the whole image
- multiple feature maps of different resolutions
- default box for each prediction
- using deep network as feature extractor
- matching strategy
- Hard negative mining - balance positive and negative training examples



Thank you for your time

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